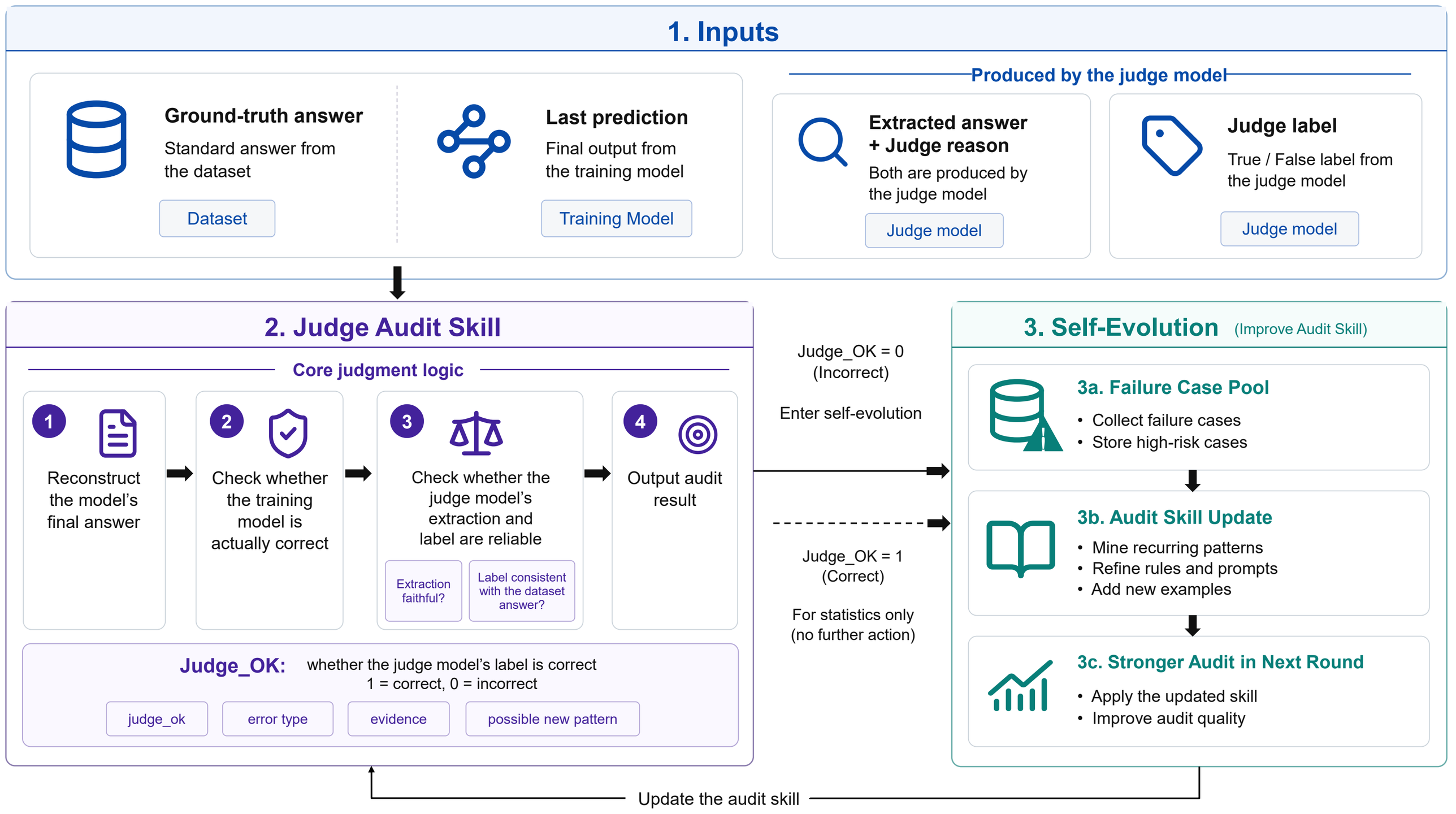

1. Inputs
Ground-truth answer
Standard answer from the dataset
Dataset
Last prediction
Final output from the training model
Training Model
Produced by the judge model
Extracted answer + Judge reason
Both are produced by the judge model
Judge model
Judge label
True / False label from the judge model
Judge model
2. Judge Audit Skill
Core judgment logic
1
Reconstruct the model's final answer
2
Check whether the training model is actually correct
3
Check whether the judge model's extraction and label are reliable
Extraction faithful?
Label consistent with the dataset answer?
4
Output audit result
Judge_OK: whether the judge model's label is correct
1 = correct, 0 = incorrect
judge_ok
error type
evidence
possible new pattern
Judge_OK = 0 (Incorrect)
Enter self-evolution
Judge_OK = 1 (Correct)
For statistics only (no further action)
3. Self-Evolution (Improve Audit Skill)
3a. Failure Case Pool
• Collect failure cases
• Store high-risk cases
3b. Audit Skill Update
• Mine recurring patterns
• Refine rules and prompts
• Add new examples
3c. Stronger Audit in Next Round
• Apply the updated skill
• Improve audit quality
Update the audit skill